

DHAVAL KAKKAD

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PROFESSIONAL SUMMARY

Mechanical Engineer and R&D leader specializing in CAD/CAM, simulation-driven product development, and medical device engineering. Currently Director – R&D at Vyom Surgicals, an early-stage medical device startup, leading 2 products through active R&D using Siemens NX, SolidWorks, ANSYS, FEA, and CFD. Combines engineering analysis with direct clinical exposure — including observation of live neurosurgical procedures — to inform device design from first principles. Pursuing M.E. (CAD/CAM) with First Class with Distinction. Published researcher with a track record of translating engineering analysis into manufacturable, validated product designs. Seeking to apply expertise in mechanical design, CAE, and research-based innovation to advanced engineering and academic roles.

CORE COMPETENCIES

CAD/CAM • Siemens NX • SolidWorks • AutoCAD • Finite Element Analysis (FEA) • Computational Fluid Dynamics (CFD) • ANSYS Mechanical • Structural Analysis • Design Validation • Design for Manufacturing (DFM) • Product Development • Mechanical Design • Engineering Optimization • Technical Documentation • Medical Device Engineering • Engineering Leadership • Research & Innovation

PROFESSIONAL EXPERIENCE

Director – Research & Development | Vyom Surgicals LLP

Aug 2025 – Present

- Lead R&D at Vyom Surgicals, an early-stage medical device startup, directing 2 products currently in active R&D through concept, design, and validation phases.
- Develop 3D CAD models and assemblies in Siemens NX and SolidWorks for both products, translating clinical requirements into manufacturable design concepts.
- Conduct FEA and CFD simulations to validate structural integrity and fluid performance ahead of physical prototyping.
- Observed 2 live neurosurgical procedures to study surgical workflow, instrument handling, and procedural constraints, directly informing device ergonomics and design requirements.
- Own prototype development and technical documentation, ensuring designs meet manufacturability and regulatory readiness standards.
- Apply simulation-driven design methodology to improve product performance, manufacturability, and reliability across the R&D pipeline.
- Contribute to strategic planning and product commercialization decisions as part of the founding leadership team.

Service Advisor Intern | Bharat Motors, Rajkot

Jan 2024 – Apr 2024

- Managed service scheduling and workflow coordination, supporting day-to-day customer maintenance planning.
- Served as liaison between customers and technical service teams, translating diagnostic findings into clear service plans.
- Gained hands-on exposure to automotive systems and maintenance procedures, reinforcing practical mechanical engineering fundamentals.

EDUCATION

Master of Engineering (M.E.) – Mechanical Engineering (CAD/CAM)

2024 – 2026

Government Engineering College, Rajkot — First Class with Distinction, CGPA 8.07

Bachelor of Engineering (B.E.) – Automobile Engineering

2019 – 2024

Government Engineering College, Rajkot — First Class, CGPA 6.88

RESEARCH PUBLICATIONS

- “Analysis of Sinking of the Titan Submersible in Accordance with Carbon Fibre: A Case Study” — International Journal of Research in Engineering and Science (IJRES)
- “A Detailed Study: Enhancing Battery Performance in Automobiles for a Sustainable Future” — International Journal of All Research Education and Scientific Methods (IJARESM)

LEADERSHIP & ACHIEVEMENTS

- National-Level 10m Pistol Shooter: Represented Gujarat Technological University (GTU) at the All India Inter-University (AIU) 2025 Shooting Championship, Chandigarh
- Naval NCC: Petty Officer Cadet (3 Gujarat Naval Unit), 'C' Certificate Holder, State-Level Firing Competition Participant, National Debate Competition Winner – Gujarat Directorate
- Managed a full Combined Annual Training Camp (CATC) for 3 Gujarat Naval Unit NCC as Senior Ex-Cadet, overseeing camp logistics, cadet coordination, and daily operations
- Best Commendation Award, Bali Asia International Model United Nations (BAIMUN) – represented Belgium in UNGA simulation among delegates from 40+ countries
- District-Level Debate and Quiz Competition winner; multiple public speaking and leadership recognitions
- Published Author, “The Whispering Voice of Leadership: A Boy’s Journey from Silence to Strength”